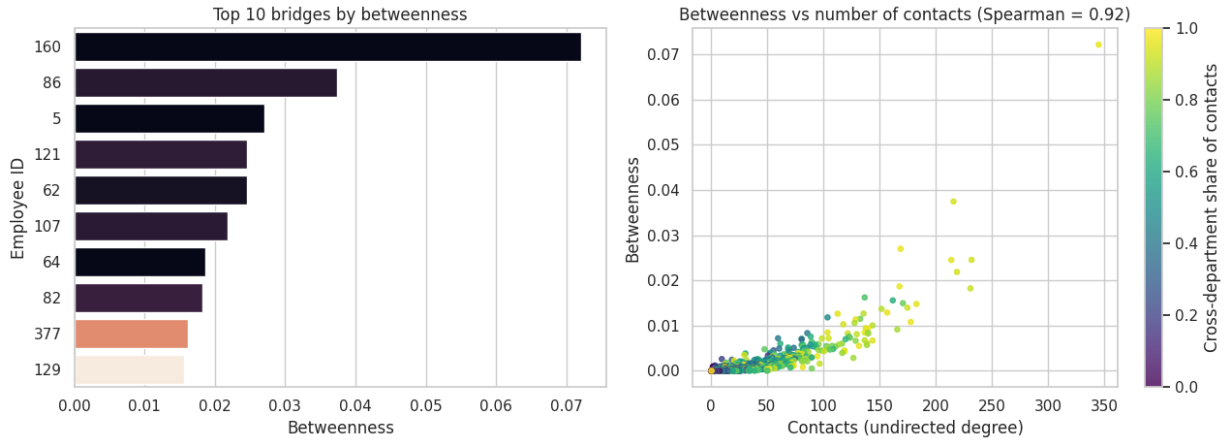
Organizational Network Intelligence

Network analysis of internal email communication

Aisha Hamza

> **Main finding.** A small group in Department 36, only 22 of 1,005 people, sits at the centre of the organisation's email network. It holds the top five receivers of email and six of the top ten bridges between groups.

Using Python, I analysed the internal email network of a large European research institution (1,005 employees and 25,571 directed email links) to find out who the key communicators are, how the organisation clusters, and where information flows or gets stuck. I built the network with NetworkX and measured it with centrality, community-detection, shortest-path, machine-learning and anomaly-detection methods, and tested whether the org chart matches how people actually communicate.



Employee 160 is the biggest bridge by a wide margin, and almost all of the top bridges contact people outside their own department.

1. Introduction

An organisation needs to know who its people rely on to get things done. Without that view, key communicators and the people who connect departments go unnoticed, and silos form without anyone seeing them. This project analyses the SNAP email-Eu-core network to answer practical questions about communication. The analysis aims to:

- Identify the key communicators by emails received and sent.
- Detect communities and test whether they match departments.
- Measure how much communication stays within departments and how much crosses them.
- Measure how efficiently information travels through the network.
- Test whether future key communicators can be predicted.
- Find the people who bridge groups and the people with unusual email behaviour.

2. Data Overview

The data comes from SNAP (snap.stanford.edu/data/email-Eu-core.html) and has two files. The network has **1,005 employees** and **25,571 directed links** across **42 departments**, with a density of 0.0253, so only about 2.5% of all possible connections are used. A link u -> v means u sent v at least one email, so links are unique sender-receiver pairs, not a count of emails.

File	Column	Description
email-Eu-core	Sender	ID of the employee who sent at least one email
email-Eu-core	Receiver	ID of the employee who received it
email-Eu-core-department-labels	Node	Employee ID
email-Eu-core-department-labels	Department	Department the employee belongs to (0 to 41)

3. Data Cleaning and Preparation

Issue	Treatment
Missing values	None found in either file
Duplicate sender-receiver rows	None found
642 self-loops (people emailing themselves)	Removed from network metrics, leaving 24,929 links. The intra-department share is reported both with and without them
Data types	Checked, all integer IDs, no changes needed

The cleaned network has 20 weakly connected components, and the largest strongly connected component holds 803 people. Shortest-path results use that component. All other metrics use the full cleaned network.

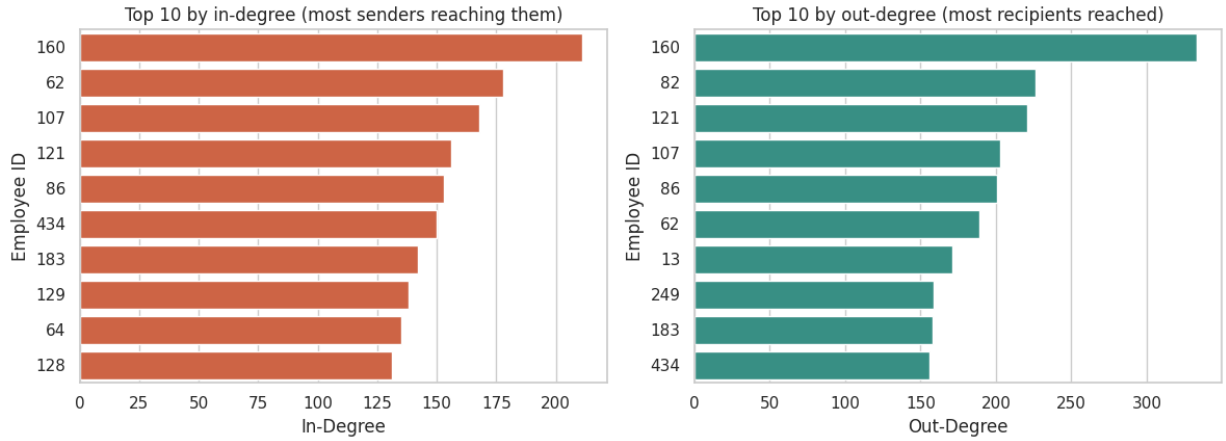
4. Analysis and Findings

4.1 Key communicators

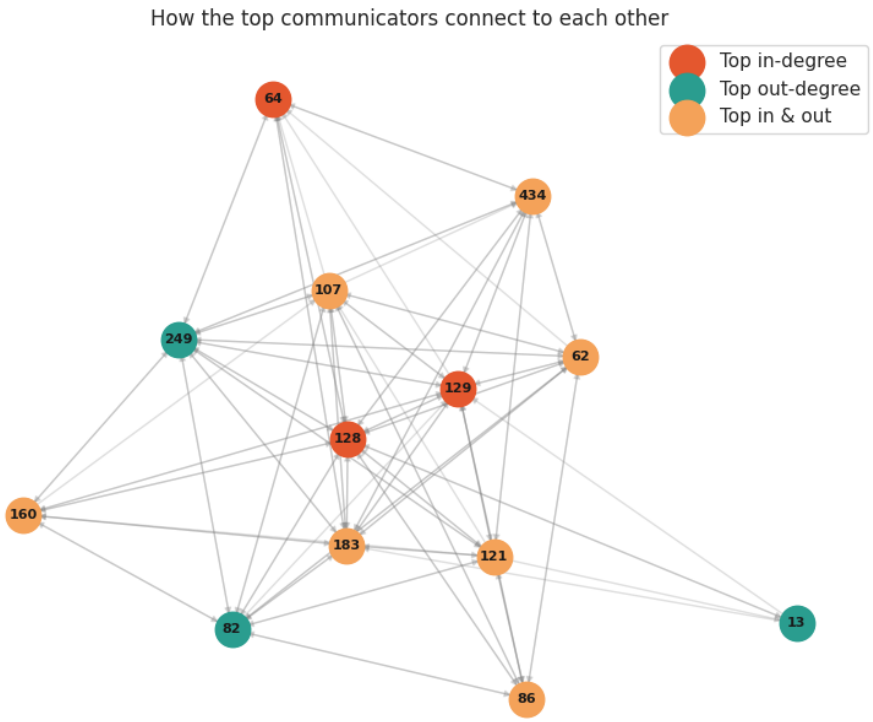
In-degree is the number of different people who emailed someone; out-degree is the number of different people they emailed. Both are measured on the full network. The median employee has an in-degree of 17 and an out-degree of 14.

Rank	Top in-degree (employee, dept, value)	Top out-degree (employee, dept, value)
1	160, dept 36, 211	160, dept 36, 333
2	62, dept 36, 178	82, dept 36, 226
3	107, dept 36, 168	121, dept 36, 221
4	121, dept 36, 156	107, dept 36, 203
5	86, dept 36, 153	86, dept 36, 201

Employee 160 leads both lists, receiving from about 12 times and emailing about 24 times as many people as the median employee. Seven employees (62, 86, 107, 121, 160, 183 and 434) appear in both top-10 lists, and in-degree and out-degree are strongly linked (Spearman 0.90), so heavy receivers are also heavy senders. All of the top five in each list are in Department 36.



Top 10 employees by in-degree and by out-degree.

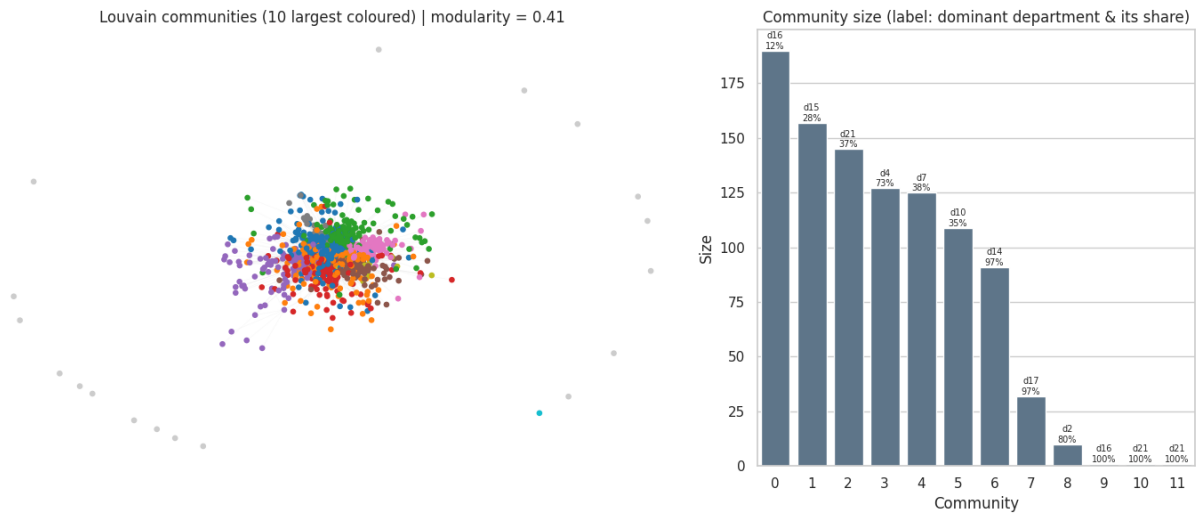


How the top communicators connect to each other.

4.2 Communities

Louvain detection found **28 communities** with a modularity of **0.41**, which means the network has clear group structure. Seven large communities of 91 to 190 people cover 944 of the 1,005 employees. Agreement with the department labels is moderate (NMI 0.60, ARI 0.35), and **43.7%** of links run between communities.

Only a few communities line up with one department. Community 6 is 97% Department 14, community 7 is 97% Department 17, and community 3 is 73% Department 4. The largest community (190 people) is a mix, with no department above 12%. So the communication groups partly follow the org chart and partly cut across it.

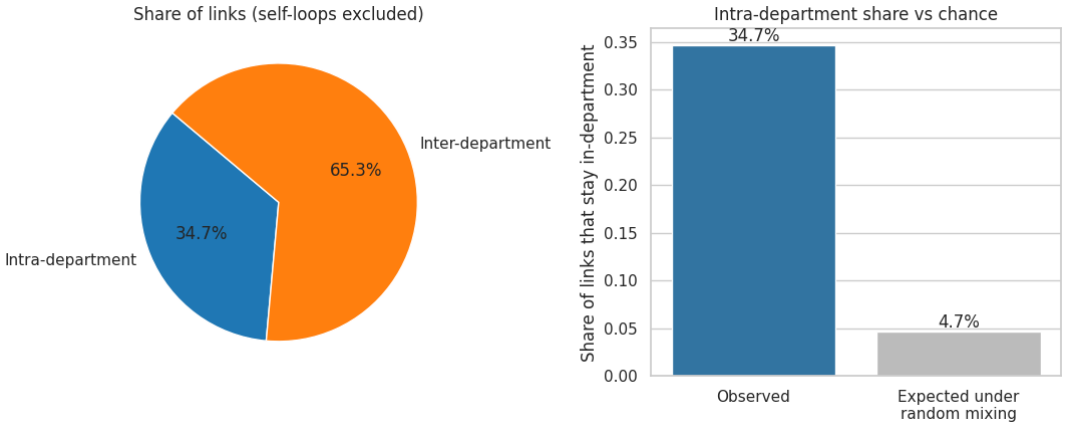


Louvain communities (left) and the size of the largest ones with their dominant department (right).

4.3 Within departments or across them

Measure	Share of links
Intra-department (including self-loops)	36.3% (9,287 links)
Intra-department (excluding self-loops)	34.7%
Expected if links were placed between random pairs	4.7%

Most links (about 65%) cross departments, so this is a cross-functional organisation. But people still talk inside their own department about **7 times more** than chance would give, so departments do matter.

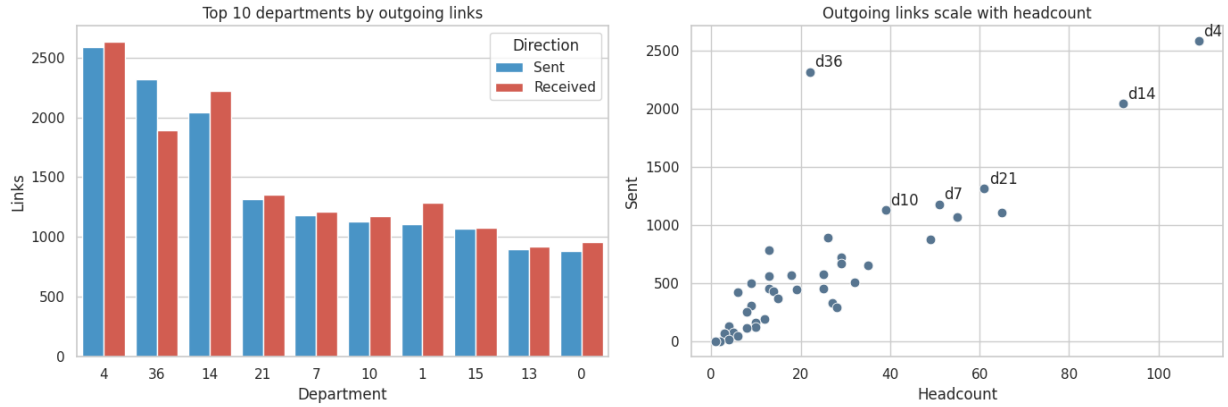


Intra vs inter-department share of links, compared with random mixing.

4.4 Department activity

Department	Headcount	Sent	Received	Sent per person
4	109	2,584	2,632	23.7
36	22	2,318	1,889	105.4
14	92	2,044	2,217	22.2
21	61	1,315	1,356	21.6
7	51	1,182	1,212	23.2

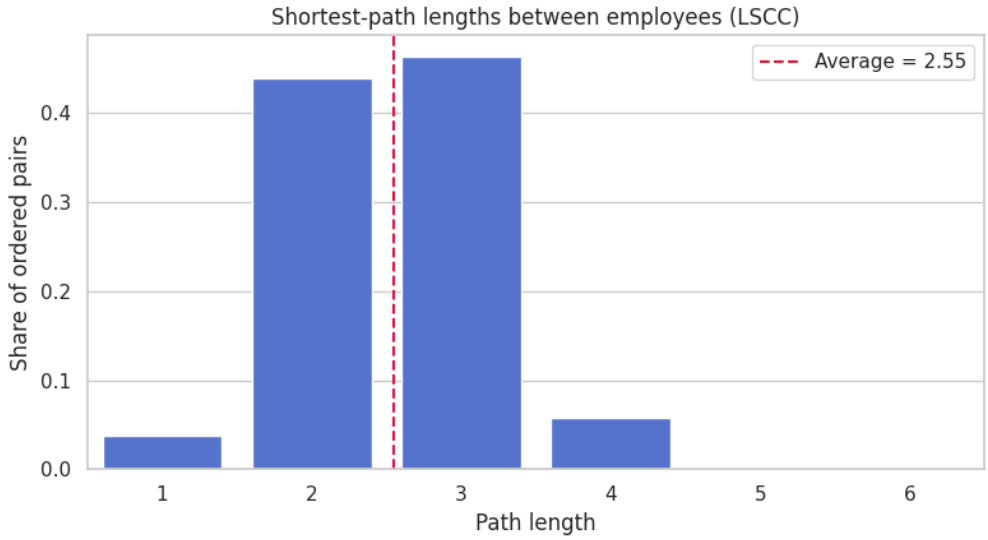
Department 4 sends the most links in total, but it is also the largest. Department 36 stands out: with just 22 people it sends 2,318 links, about 105 per person against 17 to 35 for the other top departments, and it sends 429 more links than it receives. Departments 14 and 1 receive more than they send. Outside Department 36, activity mostly follows headcount.



Top 10 departments by outgoing links, and outgoing links against headcount.

4.5 Information flow

The largest strongly connected component holds **803 people (79.9% of the network)**. The average shortest path is **2.55** steps and the diameter is **6**. About **94.1%** of pairs can reach each other within 3 hops, and 3.7% of pairs are direct contacts. Information reaches almost anyone in two or three steps, so the network is efficient. The long paths of 5 to 6 steps involve very few pairs.



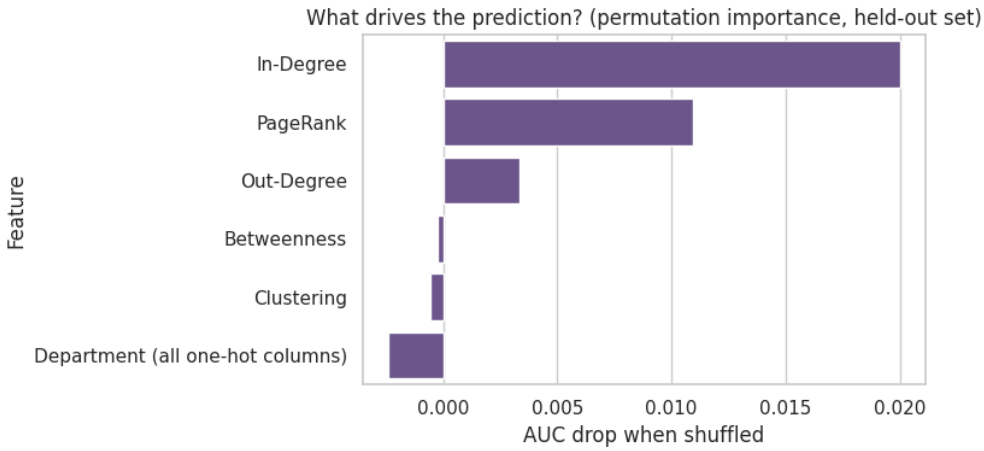
Distribution of shortest-path lengths in the largest strongly connected component.

4.6 Predicting key communicators

There are no timestamps, so I used a link-holdout test: features come from 70% of links, and the target is being in the top 10% by in-degree (105 of 1,005 employees) in the other 30%. This keeps the answer out of the inputs, unlike predicting a label such as PageRank from the features it is built from.

Model	ROC AUC	Average precision
Baseline: logistic on in-degree only	0.982	0.884
Random forest: department only	0.625	0.239
Random forest: network features	0.968	0.848
Random forest: network + department	0.979	0.883

Past in-degree is by far the best predictor. A random forest with all features does not beat the one-feature baseline (0.979 against 0.982). Department alone is only slightly better than guessing and adds nothing once network features are included. Permutation importance agrees: in-degree (0.020 AUC drop) and PageRank (0.011) matter, while department, betweenness and clustering do not.

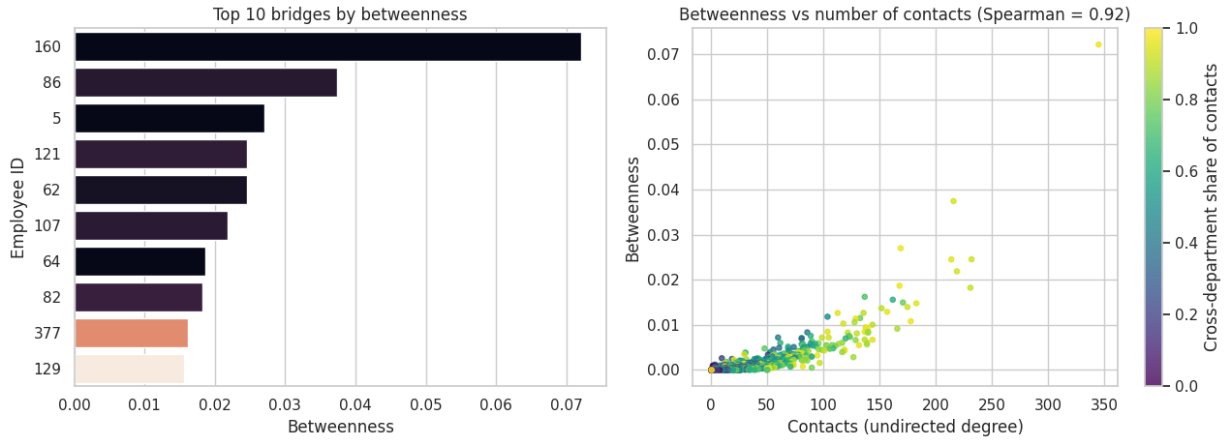


Permutation importance on a held-out test set.

4.7 Bridges

Employee	Department	Betweenness	Contacts	Cross-department share
160	36	0.072	345	96.8%
86	36	0.037	216	93.5%
5	25	0.027	169	97.0%
121	36	0.025	232	92.7%
62	36	0.025	214	95.3%

Employee 160 has almost twice the betweenness of the next bridge. Six of the top ten bridges are in Department 36 and two (5 and 64) are in Department 25. Each of the top eight has more than 90% of contacts outside their own department and reaches 31 to 39 of the 42 departments. Betweenness tracks the number of contacts closely (Spearman 0.92), and 8 of the top 10 bridges are also in the top 10 by total degree, so the best-connected people are also the ones the network depends on. The network has 73 articulation points, people whose removal cuts someone off from the rest.



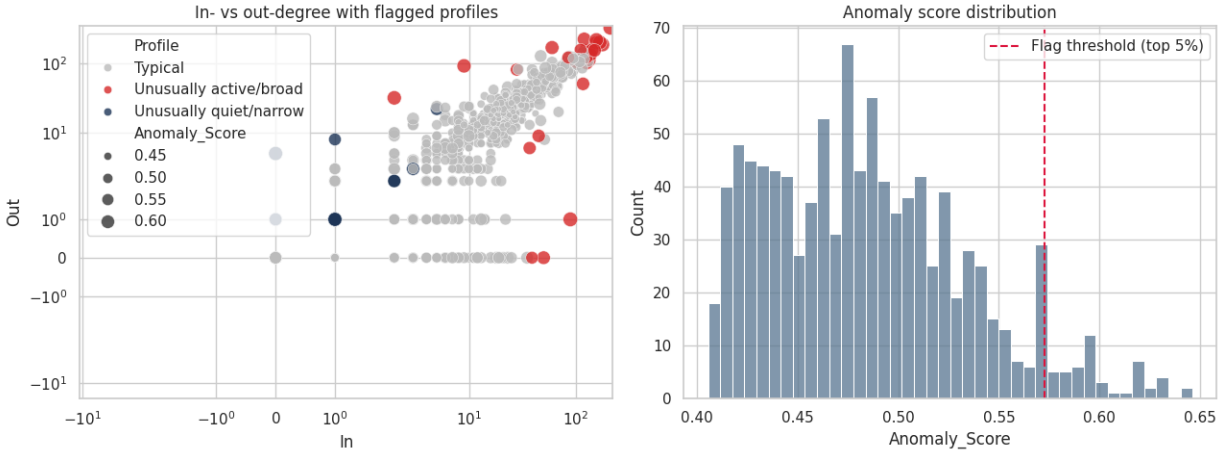
Top 10 bridges by betweenness, and betweenness against number of contacts.

4.8 Unusual email behaviour

Isolation Forest flagged **51 employees** (the top 5% most unusual profiles): 26 unusually active or broad and 25 unusually quiet or narrow.

- **Unusually active:** the top communicators 160, 107, 62 and 121 are flagged simply for being so connected. Employees 564 (2 in, 32 out) and 971 (9 in, 93 out) send widely but receive almost nothing, which looks like broadcast or announcement behaviour. Employee 365 receives from 89 people but sends to only 1.
- **Unusually quiet:** employees 763 and 837 have a single sender and a single recipient, and 979 emails 5 people but receives nothing.

A flag is a lead to look at, not evidence of a problem. Most are explained by role.



In-degree against out-degree with flagged profiles, and the anomaly score distribution.

5. Conclusion

The organisation is well connected: information travels in about 2.5 steps, most communication crosses departments, and the network has clear group structure. But the connections are concentrated in a few people.

- **Key communicators:** Employee 160 and four colleagues in Department 36 receive and send more than anyone else.
- **Communities:** 28 communities (7 large) partly follow departments but also cut across them (NMI 0.60).
- **Departments:** 65% of links cross departments, yet staying in-department is about 7 times more common than chance.
- **Information flow:** the average path is 2.55 steps, and 94% of pairs are within 3 hops.
- **Prediction:** past in-degree predicts future key communicators far better than department does.
- **Bridges and anomalies:** the main bridges are the same people as the main communicators, and most flagged profiles are explained by their role.

Overall, **a small, highly connected core in Department 36 carries a large share of the organisation's communication**.

6. Recommendations

1. **Use the key communicators for announcements and knowledge sharing.** Employees 160, 62, 107, 121 and 86 reach the most people, so they are the fastest route for organisation-wide messages.
2. **Reduce reliance on a few bridges.** Employee 160 carries nearly twice the bridging load of anyone else, and Department 36 holds most of the top bridges. Document what they connect and build second links so communication does not depend on them.
3. **Look at Department 36's workload.** Twenty-two people send about 105 links each, several times the norm, which is worth checking for overload.
4. **Review flagged profiles for a business explanation first.** Check whether 365 is a shared mailbox or list, whether 564 and 971 are broadcasters, and whether the very quiet employees such as 763, 837 and 979 are inactive accounts or people who need to be brought into communication.
5. **Add timestamps and message volumes.** Prediction and anomaly detection would be far stronger with dates and message counts.

7. Assumptions and Limitations

- The data is a public research dataset, not a live company system.
- It is a single snapshot with no timestamps, so changes over time cannot be analysed.
- Links record at least one email, so message volume and content are unknown.
- Department is one flat label, with no seniority or role information.
- Anomalies are statistical outliers, not evidence of wrongdoing.
- The prediction test is a proxy for "becoming" a key communicator, since true future behaviour is not observed.
- Community detection and the link-holdout split use a fixed random seed (42), so exact values can shift slightly with a different seed.

Tools: *Python, Pandas, NetworkX, scikit-learn, Matplotlib, Seaborn, SciPy, Google Colab.*